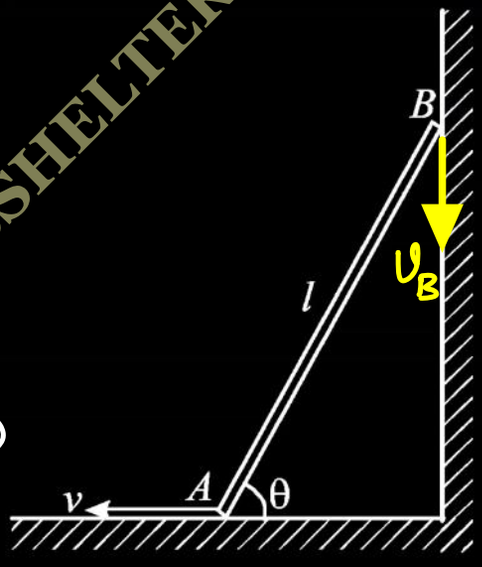
CONSTRAINT RELATIONS: In mechanical systems, motion of one part of the system requires other parts of the systems to move according to some mathematical equation. The equations governing these relationships are called constraint relations. For simple systems, we will be able to figure out constraint relations just by visual inspection. For complicated systems we may have to follow rigorous mathematical approach.

CONSTRAINT RELATION BY VISUAL INSPECTION:

EXAMPLE: Find the velocity and acceleration constraint relation between points A and B for the shown sliding ladder.



v = constant

Solution: Since ladder is a rigid body, velocity of approach of any two particles on it must be zero. Thus: $v\cos\theta = U_B \sin\theta$ Ans. (1)

Now differentiating velocity constraint:

$$-v\sin\theta\left(d\theta/dt\right) = U_B\cos\theta\frac{d\theta}{dt} + a_B\sin\theta$$

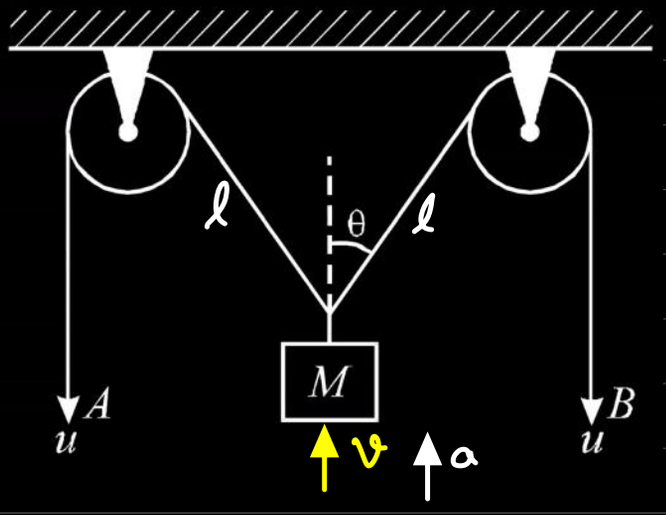
Thus $$a_B = -\frac{d\theta}{dt}\left(v\sin\theta + U_B\cos\theta\right)\times\frac{1}{\sin\theta} \quad (2)$$

$$\frac{d\theta}{dt} = -\left(\frac{v\sin\theta + U_B\cos\theta}{l}\right)$$ (∵ Recall ω defined while discussing velocity of approach).

Thus knowing U_B and $\frac{d\theta}{dt}$ we can find a_B by substituting them in equation (2)

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EXAMPLE: For the shown system, A and B are being pulled with uniform velocities. Find (a) constraint relation between v and u (b) Instantaneous acceleration of M. Treat pulleys like point pulleys.



Solution: Since length of the strings is constant, approach velocity of M towards right pulley must be equal to separation velocity of B from that pulley. Thus:

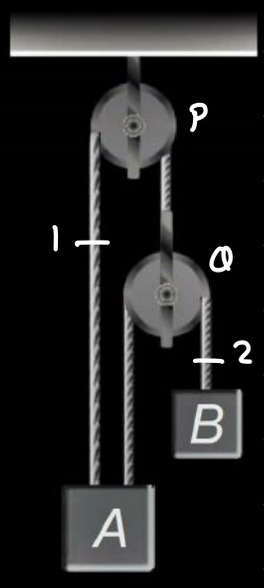
$$v\cos\theta = u \quad (1) \text{ ans.}$$

Now differentiating (1), $\quad a\cos\theta - v\sin\theta \frac{d\theta}{dt} = 0 \quad (2)$

Also $d\theta/dt = v\sin\theta/l \quad (3)$ Now substituting for v from (1) and $d\theta/dt$ from (3) in (2) will give acceleration.

Blunder Alert! Just because $v\cos\theta = u$ does not imply $a\cos\theta = du/dt$ ∵ θ is also changing with time.

EXAMPLE: For the shown Atwood machine find relation between velocities of blocks A and B



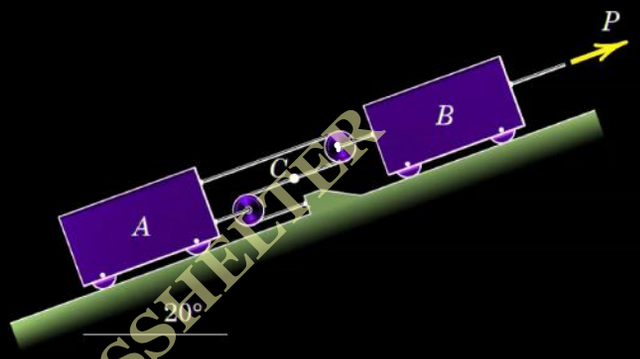
Solution: For applying visual inspection method, we slacken and tighten the strings one by one. Now consider following operations:

1. Move A by 1 unit (All other pulley and blocks unmoved) this creates a slack of 1 unit in both strings.
2. Now move pulley Q down to tighten string 1 (other blocks/pulleys unmove). This brings pulley Q down by 1 unit creating 2 units of additional slack in string 2 on either side of pulley Q. Thus now cumulative slack in string 2 is 3 units.
3. Pull block B down by 3 units to absorb all the slack in string 2. Now all strings are taut once again.

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In the whole process A moved up by 1 unit while B moved down by 3 units. Thus their displacements are in 1:3 ratio and so also their velocities an accelerations / all are moving along single straight lines without any angles / angular velocities coming in to picture.).

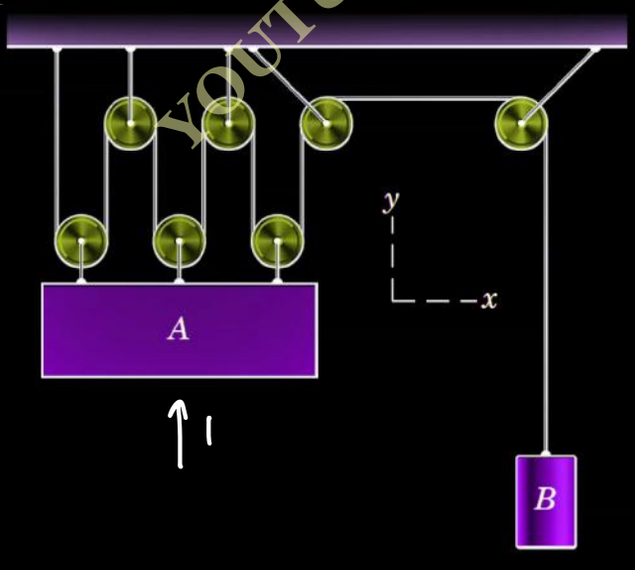
EXAMPLE: Find a relation between velocities of A and B for shown adjacent figure.



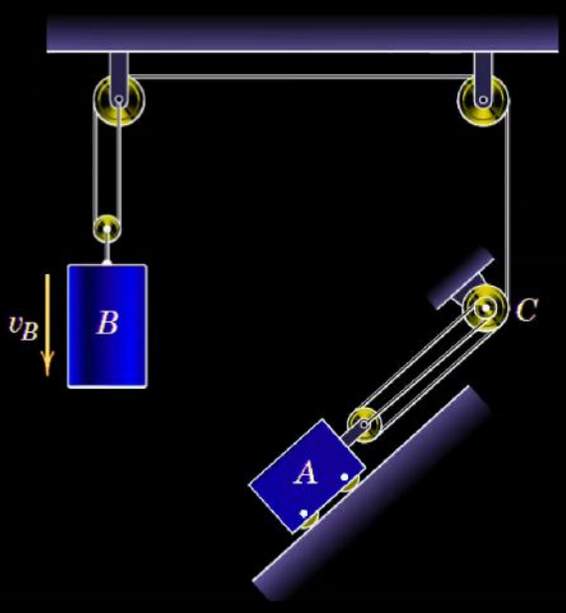
Solution: Moving B 1 unit down creates 2 units of slack which means moving B 2 units upwards will consume 2 units of slack. Similarly Moving A 1 unit upwards creates 3 units of slack To absorb this slack, B will have to move upwards 3/2 units (To consume 2 units of slack 1 unit movement is required so to consume 3 units, we need 3/2 units movement, ∵ unitary method). (While creating / absorbing the slack we move one object at a time keeping everything else unmoved).

Think Yourself: Figure out constraint relations:

(a)

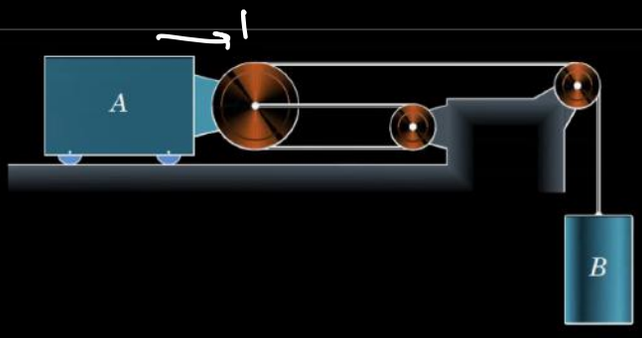


(b)

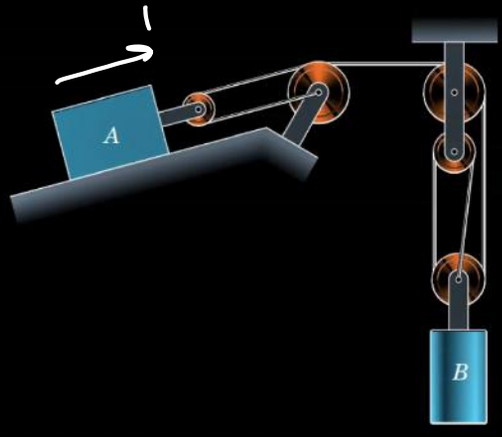


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(c)



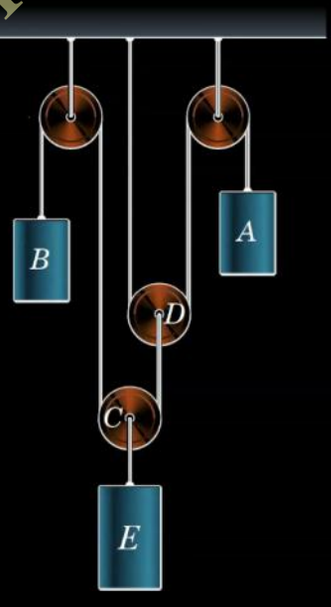
(d)



Ans: (a) $v_B = 6v_A$ (c) $v_B = 3v_A$

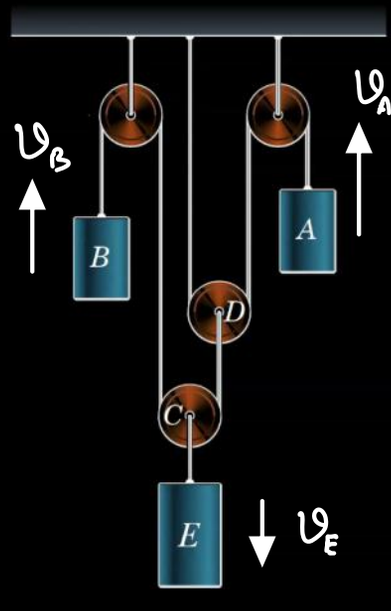
(b) $v_B = \frac{3}{2}v_A$ (d) $v_B = \frac{2}{3}v_A$

Visual Inspection With Superposition:



There can be systems with two or more degrees of freedom (A two degree of freedom system means that to comment on motion of a particular block, we need two informations about the displacements in the system). For instance in adjacent figure, displacement of E depends simultaneously on displacements of A and B. In such cases we can find the effect of individual movements, keeping the other block stationary and then superpose the effect of two motions to get the net effect. This will be more clear from following example:

Example: For the shown system, find the relation between velocities v_A, v_B and v_E



Solution: First we will move B keeping A stationary and see its effect on movement of E and then we will move A keeping B stationary to find effect on E. If we move B by 1 unit upwards, E must go down by 1/2 unit, thus if B goes up by v_B, E goes down with $v_B/2$. Now if we move A upwards by 1 unit, D moves down by 1/2 unit and then C moves down by 1/4 unit and so also E. Thus if A moves with v_A,

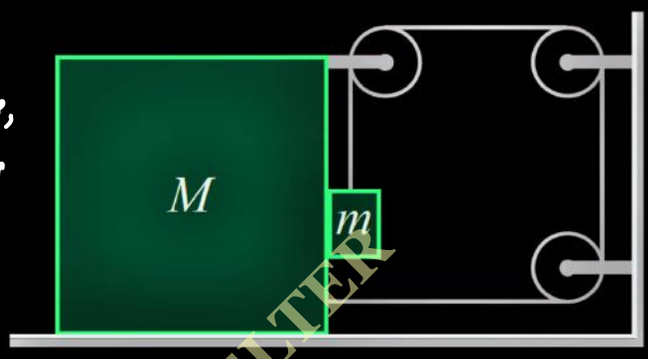
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E moves with $U_A/4$. Thus superposing now we have the complete constraint relation:

$$U_E = U_B/2 + U_A/4 \text{ ans.}$$

Visual Inspection With Simultaneous Displacements:

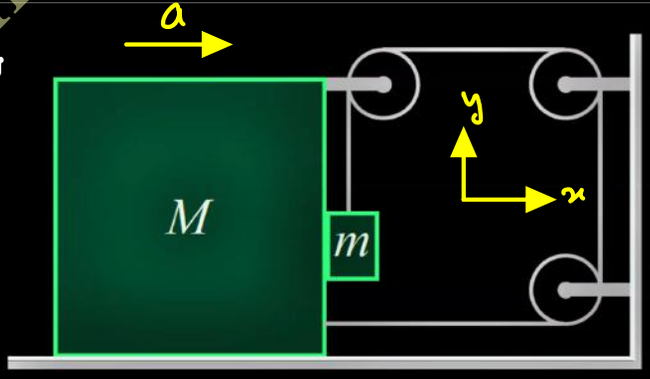
Sometimes due to nature of connections in the mechanical systems, it might not be possible to move one body at a time, still we can account for the simultaneous movements by carefully visualising what is going on. For instance in adjacent figure, moving M to the right requires simultaneous shifting of m also rightwards. The method of dealing with such systems will be clear from following example.



Example: For the shown system find acceleration of m if M is accelerating with $a\hat{i}$.

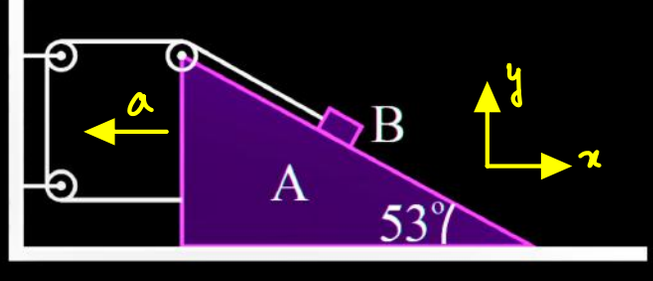
Solution: Moving M 1 unit rightwards, moves m also 1 unit rightwards and creates a slack of 2 units. To absorb this slack, m must be moved 2 units downwards to make the string taut again. Now looking at the ratios of movements we can write the acceleration of m as:

$$a_m = a\hat{i} - 2a\hat{j} \text{ ans.}$$



Example: For the adjacent figure, find acceleration of B if wedge is accelerated with $-a\hat{i}$.

Solution: If we move A and B together 1 unit left, we generate a slack of 2 units and after this B must slide down 2 units relative to A to make the string taut again.

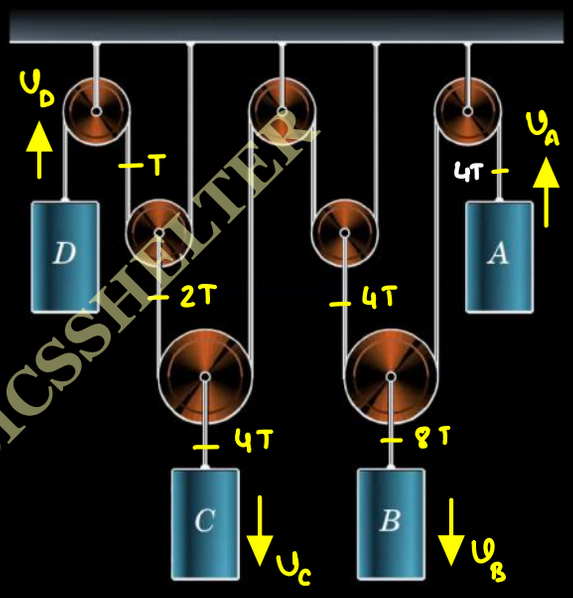


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Thus $a_B = -a\hat{i} + 2a[\cos 53^\circ \hat{i} - \sin 53^\circ \hat{j}] = \frac{1}{5}a\hat{i} - \frac{8}{5}a\hat{j}$ Ans.

TENSION POWER METHOD: In this method, if we have massless ideal strings passing over frictionless surfaces then, net sum of power developed by tension forces on all the massive objects in the system is zero.

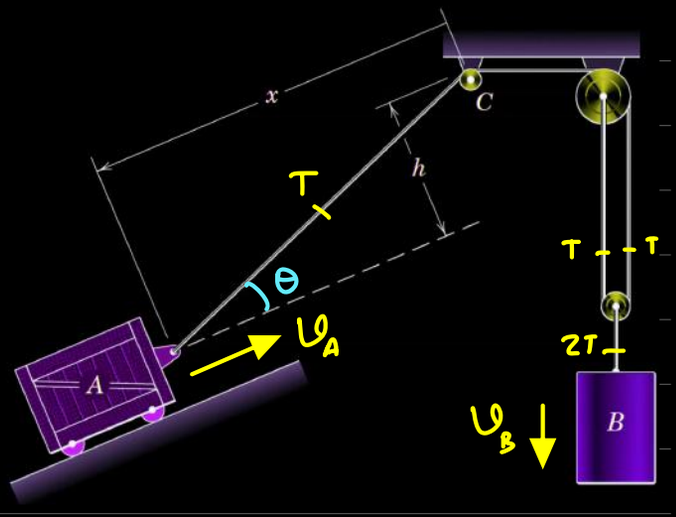
EXAMPLE: Find the constraint relation between velocities v_A, v_B, v_C and v_D for the shown system.



Solution: For massless pulleys, the net force must be zero, this gives us tensions in various threads as shown in the diagram.

Now recall that power $P = \vec{F}\cdot\vec{v}$, thus it is positive when angle between force and velocities is acute and negative if it is obtuse. Now apply $\Sigma \vec{T}\cdot\vec{v} = 0$ to the system: $4Tv_A - 8Tv_B - 4Tv_C + Tv_D = 0 \Rightarrow 4v_A + v_D = 8v_B + 4v_C$ Ans.

EXAMPLE: For the shown system, find the velocity constraint and hence also find the acceleration constraint.



Solution: $\cos\theta = x/\sqrt{x^2+h^2}$

Now $Tv_A\cos\theta - 2Tv_B = 0$

$\Rightarrow v_B = \frac{1}{2}v_A\frac{x}{\sqrt{x^2+h^2}} = \frac{1}{2}v_A\cos\theta$

Now differentiating wrt time on both sides:

$$a_B = \frac{1}{2}\left[a_A\cos\theta + v_A[-\sin\theta]\cdot\frac{d\theta}{dt}\right], \text{ Now } \frac{d\theta}{dt} = \frac{v_A\sin\theta}{\sqrt{x^2+h^2}} = v_A\frac{h}{x^2+h^2}$$

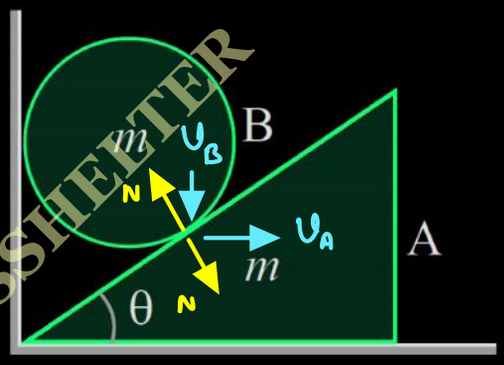
Thus $a_B = \frac{1}{2}\left[\frac{a_A x}{\sqrt{x^2+h^2}} - \frac{v_A^2 h^2}{(x^2+h^2)^{3/2}}\right]$ Ans.

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→ Tension power method must be always applied in inertial frames

→ Sometimes some bodies are not clearly shown in the figure (for instance a hand pulling a string might not be shown and instead just a pulling force might be shown. In that case we must imagine the pulling agency as a massive body and consider the power of tension on that also).

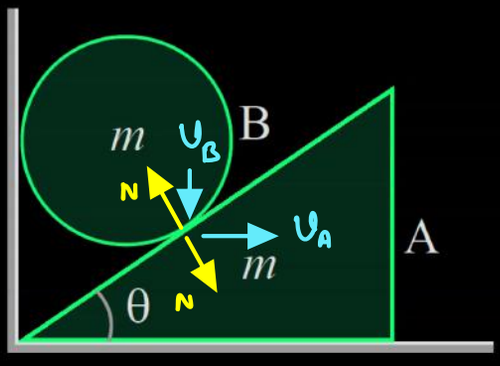
NORMAL POWER METHOD: Whenever there is a normal reaction pair at interface of two contacting surfaces, the combined power developed by the Normal reaction pair on the two surfaces is zero. (∵ velocity of approach of two contacting particles is zero, so their velocity components along the Normal are equal). Another way of applying this principle is just to equate the components of velocities along the Normal for both contacting particles.



EXAMPLE: For the shown system, find relation between v_A and v_B.

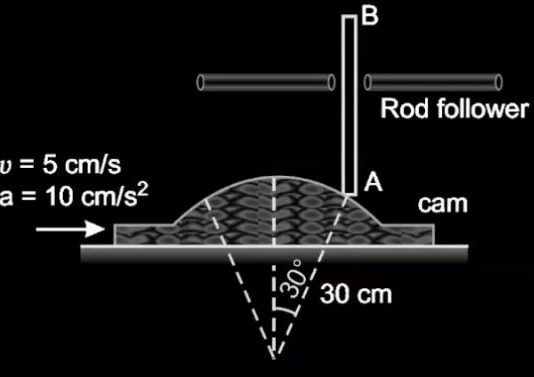
Solution: Equating the velocity components along the Normal, $v_B \cos\theta = v_A \sin\theta$

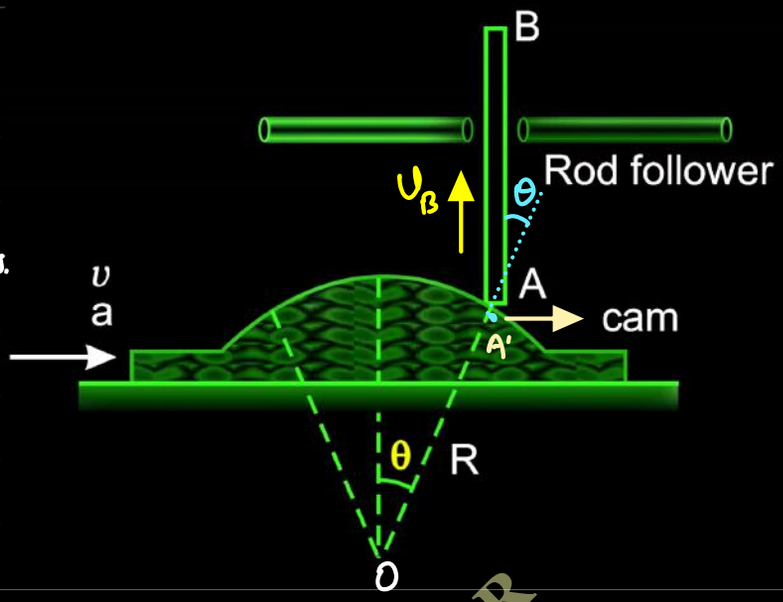
Thus $v_B = v_A \tan\theta$ Ans.



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Example: A rod follower *AB* is subjected to a vertical up and down movement while resting on the circular contour of radius 30 cm of a cam. The cam moves to the right with a velocity of 5 cm/s and an acceleration of 10 cm/s^2. Find the velocity and acceleration of point *B* on the rod at the instant of interest as shown in figure.





Solution: Equating velocity components along normal, $V_B \cos\theta = v \sin\theta$

$\Rightarrow \quad V_B = v \tan\theta \quad (1)$

$\Rightarrow V_B = 5\,cms^{-1} \times \tan 30^\circ = 5/\sqrt{3}\ cms^{-1}$ Ans.

$\approx 2.88\ cms^{-1}$

Now differentiating (1) wrt time:

$$a_B = a\tan\theta + v\sec^2\theta \frac{d\theta}{dt}$$

Now θ is the angle between line joining centre of curvature to the contact point of the rod. As the Cam moves θ also moves. Now $d\theta/dt$ can be found using the kinematics idea of angular velocity of line joining two particles. Thus,

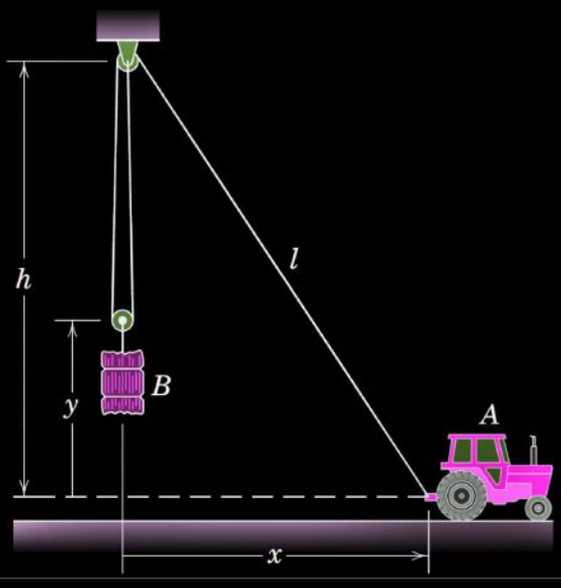
$$\frac{d\theta}{dt} = -V_B \sin\theta/R + (-v\cos\theta)/R$$

$$= -\frac{5}{\sqrt{3}}\ cms^{-1} \times \frac{1}{2} \times \frac{1}{30\,cm} - 5\,cms^{-1} \times \frac{\sqrt{3}}{2} \times \frac{1}{30\,cm} = \frac{-1}{3\sqrt{3}}\ rad\,s^{-1}$$

thus $a_B = 10\,cms^{-2} \times \frac{1}{\sqrt{3}} + 5\,cms^{-1} \times \frac{4}{3} \times \frac{-1}{3\sqrt{3}}\,\hat{j} \approx 4.5\,cms^{-2}$ Ans.

Coordinate Constraint Method:

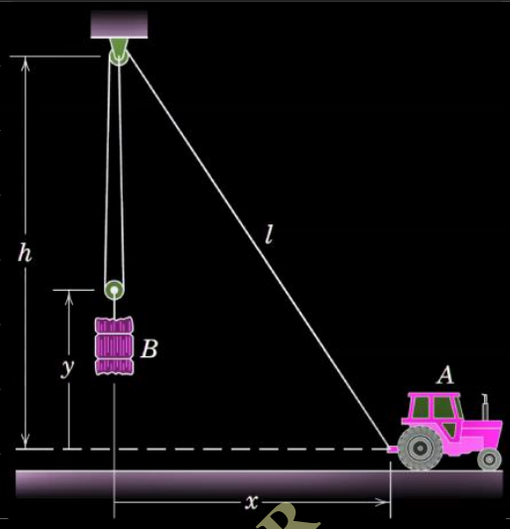
In the problems involving constraint relations, it is possible to express a geometrically conserved relation (such as length of string or a rod) in terms of coordinates of the moving points. Once we express the relation in terms of coordinates, we can differentiate the relations in coordinates to get the relations in velocities and accelerations.



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EXAMPLE:

In the shown system, the truck is moving rightwards with a constant speed v. Find a relation for upward velocity of B.



Solution: Consider the coordinates of B and A as shown in the figure. As the motion progresses, the length of the string remains constant. Thus we have the constraint:

$$2(h-y) + \sqrt{h^2+x^2} = \text{Constant}.$$

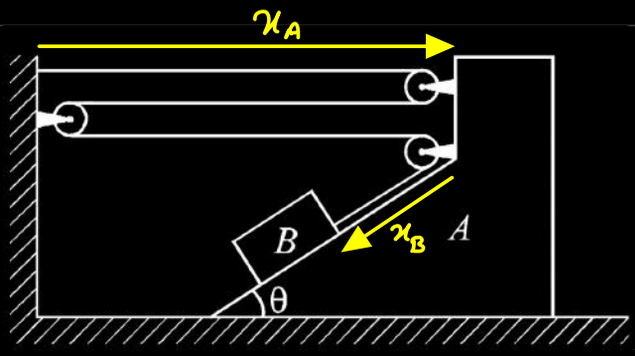
Now differentiating wrt time:

$$-2\dot{y} + \frac{1}{2\sqrt{h^2+x^2}} \cdot 2x\dot{x} = 0 \quad \Rightarrow v_B = \dot{y} = \frac{xv}{\sqrt{h^2+x^2}} \quad \text{Ans.}$$

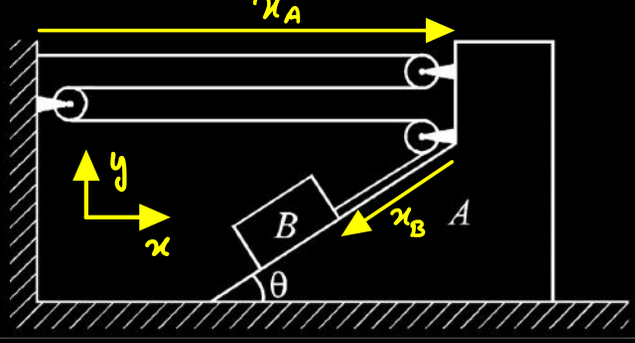
(Now if we want, we can differentiate it once more to get the constraint relation in accelerations).

CONSTRAINT WITH RELATIVE COORDINATES:

Sometimes it may be convenient to show the coordinate of a moving body relative to another moving body. In such cases it must be kept in mind that derivative of such a coordinate will give relative velocity and not the velocity in the ground frame. For instance in the adjacent figure, $\dot{x}_B$ represents velocity of B relative to A while, $\dot{x}_A$ represents velocity of A in ground frame.



EXAMPLE: In the shown system, the wedge is moving leftwards with a velocity $-v\hat{i}$. What is the velocity of B from ground frame.



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Solution: $3x_A + x_B = \text{Constt} \Rightarrow 3v_A + v_{BA} = 0 \Rightarrow v_{BA} = -3v_A$

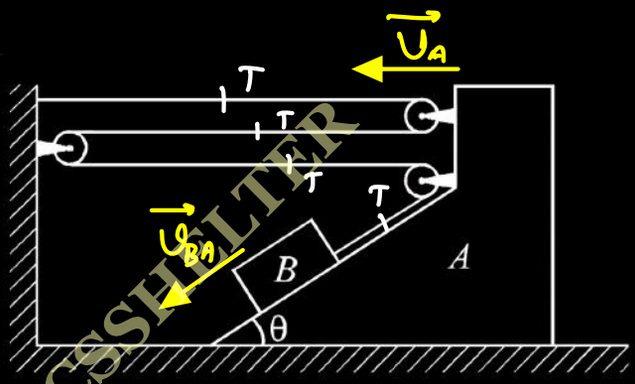
$\Rightarrow v_{BA} = +3v$ (+ve means along the coordinate, -ve means against the coordinate).

Thus:

$$\vec{v}_B = \vec{v}_A + \vec{v}_{BA} = -v\hat{i} - 3v\cos\theta\,\hat{i} - 3v\sin\theta\,\hat{j} \quad \text{i.e}$$

$$\vec{v}_B = -v\left[(1+3\cos\theta)\hat{i} + 3\sin\theta\,\hat{j}\right] \quad \text{Ans.}$$

EXAMPLE: Find the relation between v_A and v_{BA} using tension power method.



Solution: Here we must remember to not mix-up the frames. All powers must be written from the same frame, though the frame can be fixed as per our choice. So let us try first using ground frame.

Remember that $\vec{T}\cdot\vec{v}$ can always be written as $Tv_{\parallel}$ where $v_{\parallel}$ is component of velocity $\parallel$ to T (-ve if Anti-parallel). Thus working from ground frame: $(3T + T\cos\theta) \times v_A - T(v_{BA} + v_A\cos\theta) = 0$

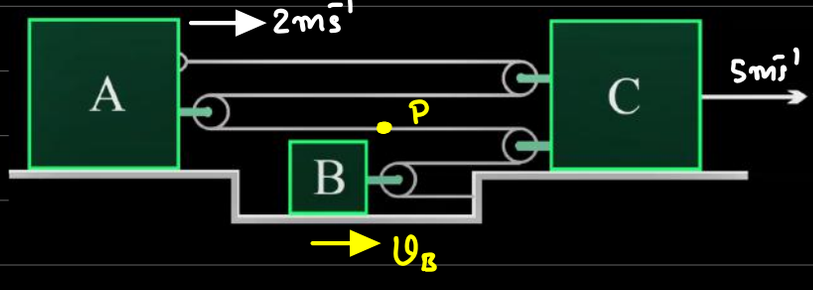
$\Rightarrow v_{BA} = 3v_A$ Ans. (Carefully understand the logic of second term above. We need to consider velocity of B also from ground itself).

→ Alternatively imagine a frame moving with velocity v_A to the left. In such a frame, wedge appears stationary however wall appears to move rightwards with speed v_A. Thus now we will need to consider work done by tension on the wall. Thus:

$3Tv_A - Tv_{BA} = 0 \Rightarrow v_{BA} = 3v_A$ (This was easier!!).

EXAMPLE:

In the adjacent figure find (a) speed of block B (b) speed of point P of the string.

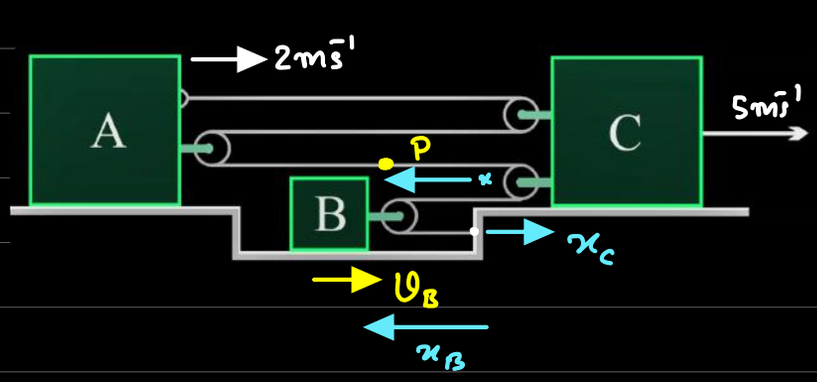


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Solution: Using tension work method,

$3T \times 2 - 4T \times 5 + 2T \times v_B = 0$

$\Rightarrow v_B = 7\,ms^{-1}$, Now considering the length of the string from fixed support to P:

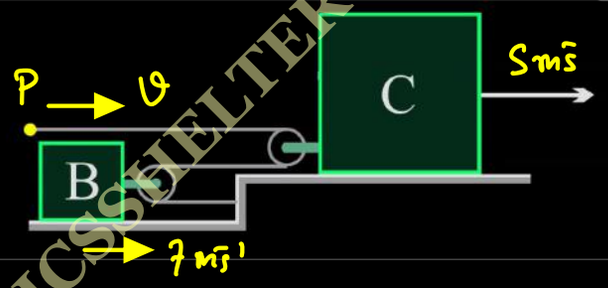


$$x + 2x_C + 2x_B = C$$

$\Rightarrow \dot{x} + 2\dot{x}_C + 2\dot{x}_B = 0 \Rightarrow \dot{x} + 2\times 5 - 14 = 0 \Rightarrow \dot{x} = +4$

Thus P is moving leftwards at $4\,ms^{-1}$.

Method II: We dropped the portion of string from P to the upper contact with A, ∵ portion of string from lower contact to P is also constant, We imagine a massive object connected to P and keep moving B and C as it is. Thus applying Tension Power method:



$$Tv + 2T \times 7 - 2T \times 5 = 0 \Rightarrow v = -4ms^{-1} \Rightarrow$$ P moves leftwards at $4ms^{-1}$ Ans. [We could have similarly made equation by dropping part of the string below P and keeping the upper part].

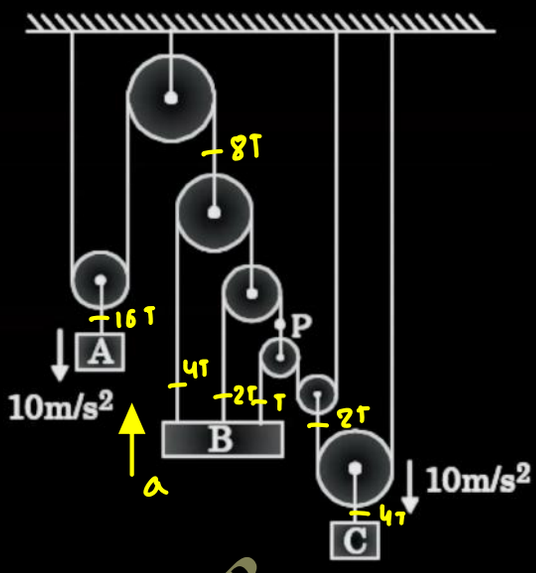
Method III: Working from frame of C, B approaches at $2ms^{-1}$ and support moves leftwards at $5ms^{-1}$. Thus P must move at $5+2\times 2\,ms^{-1}$ leftwards i.e $9\,ms^{-1}$ leftwards, thus from ground frame, velocity of P is $9-5 = 4ms^{-1}$ leftwards.

Method IV: Velocity of any pulley is average of velocities of string particles on its either sides. Now for lower pulley at C,

$$\frac{14+v}{2} = 5ms^{-1} \Rightarrow v = -4ms^{-1}$$ i.e $4ms^{-1}$ leftwards.

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EXAMPLE: In the adjacent figure, what is the acceleration of point P?



Solution: By visual inspection, the second pulley from right is moving at $20\ m/s^2$, thus string on its left side must be moving at $40\ m/s^2$ downwards. Now applying tension power method,

$$7Ta - 16T \times 10 - 4T \times 10 = 0$$

$$\Rightarrow a = \frac{200}{7}\ m/s^2.$$

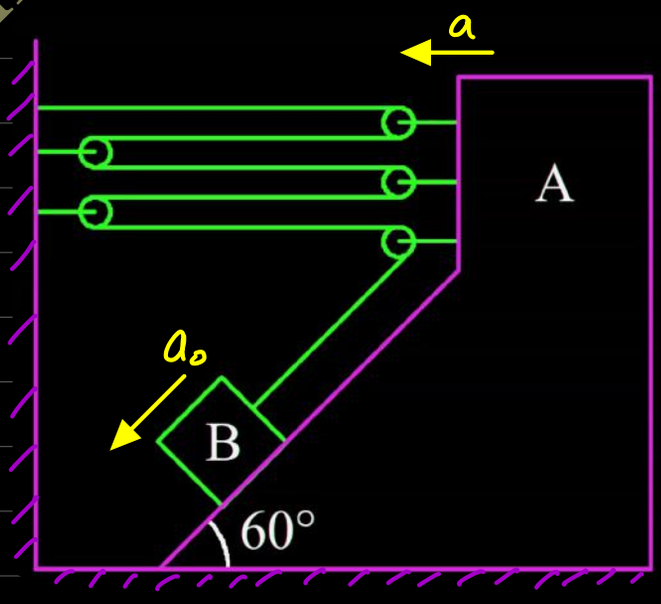
This must also be upward acceleration of string on the left side of pulley below P. Thus acceleration of this pulley = average of acceleration of particles on its either sides i.e $\left(40\downarrow + \frac{200}{7}\uparrow\right)/2 = \frac{40}{7}\ m/s^2\downarrow$. Ans.

EXAMPLE: In the shown system, Both A and B have same mass m. Pulleys, strings and surfaces are ideal. Find

(a) Acceleration a of A as seen from ground.

(b) Acceleration a_0 of B as seen from A.

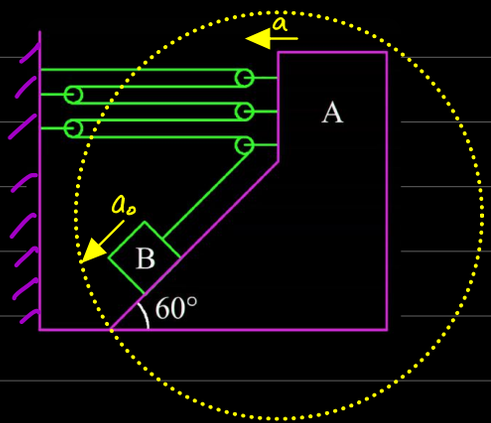
(c) String tension.



Solution: By constraint relation:

$$a_0 = 5a \quad (1)$$

Also $\Sigma F_{x_i} = \Sigma m_i a_{x_i}$, Applying this to the circled system:



$$5T = ma + m[a + 5a\cos 60°] = \frac{9}{2}ma \quad (1)$$

Now applying LOM to B in wedge frame:

$$mg\sin 60° - ma\cos 60° - T = 5ma$$

$$\Rightarrow mg\frac{\sqrt{3}}{2} - T = \frac{11}{2}ma \quad (2)$$

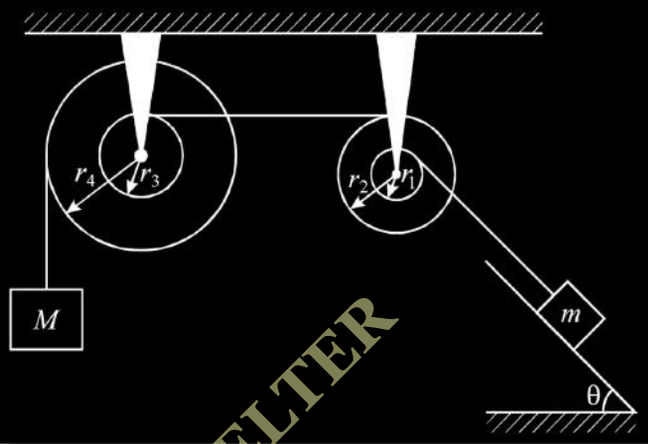
Solving (1) & (2),

$T = 9\sqrt{3}mg/128$ $\quad a = 5\sqrt{3}g/64$ and $a_0 = 5a = 25\sqrt{3}g/64$ Ans.

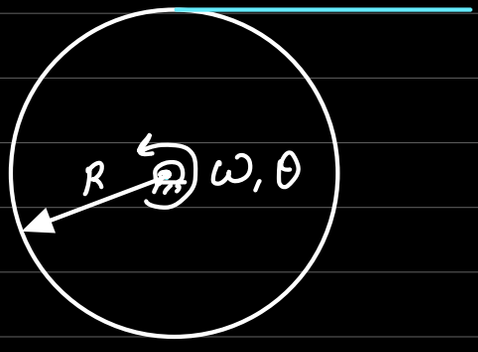
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→ Notice how by a careful choice of system, we avoided bringing Normal reaction in to equations and bypassed the unnecessary mess in forming and solving the equations).

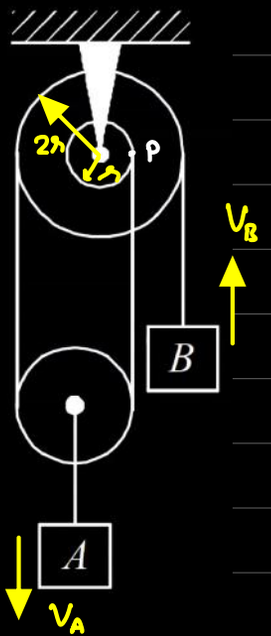
Spool / Stepped Pulley:



Sometimes, instead of total length of the straight segments in pulley blocks systems being conserved, it may increase or decrease due to thread being wound or unwound on spools or stepped pulleys (for instance consider the system shown in adjacent figure). If such a spool rotates through an angle θ, then length of the thread wound on it in the process is $l = R\theta$ thus rate of winding the thread is $\dot{l} = R\dot{\theta} = R\omega$. This information can now be used for forming the constraint equations with systems involving stepped pulleys and spools. Having understood this, we can either use the string length conservation or Tension work method to work out the required constraint.



Example: For the stepped pulley shown in adjacent figure, find constraint relations between velocity of A and B. The radii of steps are in the ratio 1:2.



Solution. Rate of string passing on outer radius, $\omega \times 2r = v_B$. Now using tension Power method:

$$T v_B - 2T v_A - T\omega r = 0 \quad \text{(Think of P as a massive particle moving up).}$$

$$\Rightarrow v_B = 2v_A + \omega r = 2v_A + v_B/2$$

$$\Rightarrow v_B = 4v_A \text{ Ans.}$$

Method 2: Visual inspection method:

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If x amount of string passes on outer radius (Moving B without moving A) then $x/2$ gets wound (due to half radius of inner step) Thus net slack created is $x/2$. This must be consumed by moving A. Thus A moves $x/4$ units ($x/4$ on both sides of pulley connected to A consumes a length $x/4$). Thus $v_A = v_B/4$.

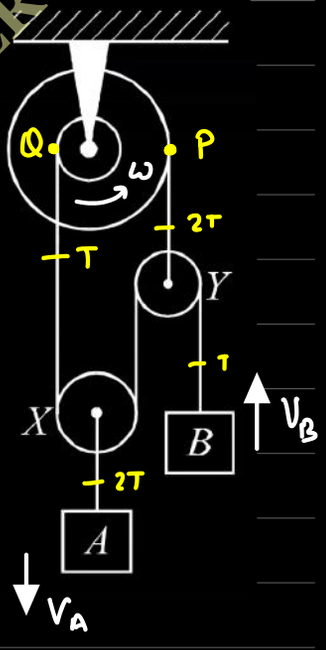
→ For easy visualization think of tangent points on stepped pulleys as being massive object connection points for the strings.

EXAMPLE: In the shown system, step radii are R and 3R. Find constraint relation between ω, v_A and v_B.

Solution: Imagining massive particles at P and Q with speeds $3\omega R$ and ωR respectively, applying tension work method:

$$T v_B - 2T \times 3\omega R + T\omega R - 2T v_A = 0$$

$$\Rightarrow \quad v_B = 2v_A + 5\omega R \quad \text{Ans.}$$



EXAMPLE:

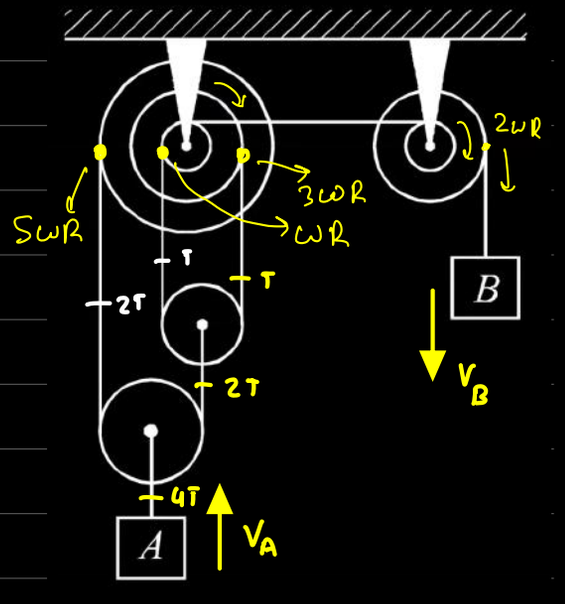
In the shown system, for first pulley, step radii are R, 3R and 5R while for second pulley they are R and 2R. Find constraint relation between v_A and v_B

Solution: Velocities of contact points at vertical tangents of the pulleys are shown. Now applying Tension Power method:

$$4T v_A - 2T \times 5\omega R - T\omega R + T \times 3\omega R = 0$$

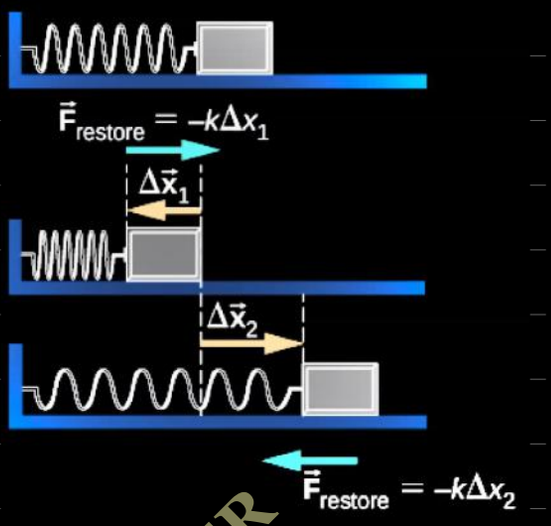
$$\Rightarrow \quad 4v_A = 8\omega R = 4v_B$$

Thus $v_B = v_A$ Ans.



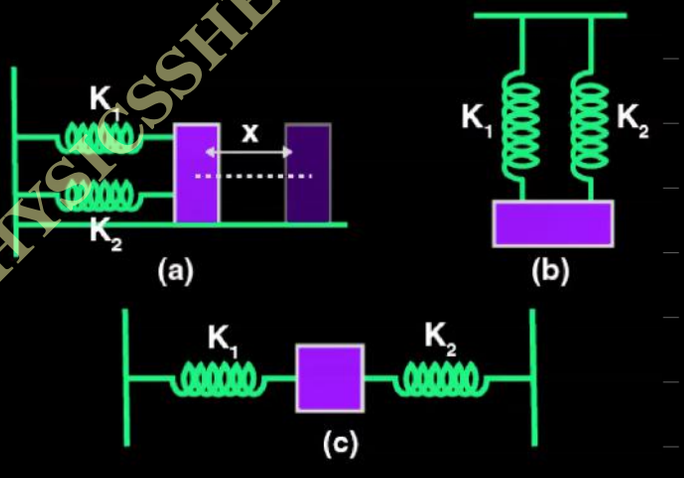
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Spring Force: Whenever a spring is compressed or extended from its natural length, it applies a restoring force on the pulling agent which is proportional to the change in length. Mathematically $\vec{F} = -k\Delta\vec{x}$ where Δx is the deformation in the spring from its natural length.



Springs In Parallel:

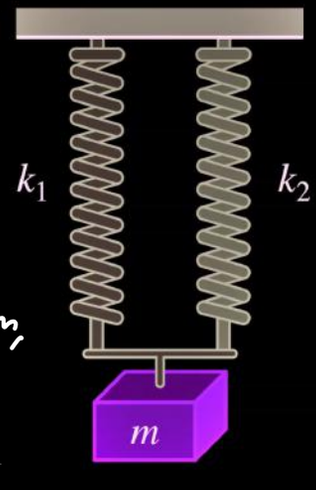
When by virtue of configuration, if two or more springs deform by the same amount while sharing the burden of a load, we call the springs to be connected in parallel. In parallel, the forces offered by the springs get simply added up. Springs in parallel sometimes might not look to be in parallel (for instance see the figure c above. When we displace the block by a distance x to the right, the force experience by it due to springs is $k_1 x + k_2 x$ just as in case (a), so we deem that it is a case of parallel connection).



Effective Spring Constant And Load Share Of Parallel Springs:

Let a parallel combination of springs extend by a distance x, then the net force offered by the system is $F = F_1 + F_2 = k_1 x + k_2 x = (k_1 + k_2)x$. Thus the system behaves as if it were a single spring of constant $k_{eff} = k_1 + k_2$. Thus for parallel combination,

$$\boxed{k_{eff} = k_1 + k_2 + \dots}$$



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→ Note that in parallel combination, force offered by each spring is proportional to its spring constant. Thus, if F is the total load shared by the combination, individual load shares are given as:

$$F_1 = \left(\frac{k_1}{k_1 + k_2}\right) F \quad \text{and} \quad F_2 = \left(\frac{k_2}{k_1 + k_2}\right) F.$$

for the case of two springs (similar formula can be written for case of multiple springs).

SPRINGS IN SERIES:

Two springs which share a portion of net displacement and carry the same load by virtue of configuration are said to be in series.



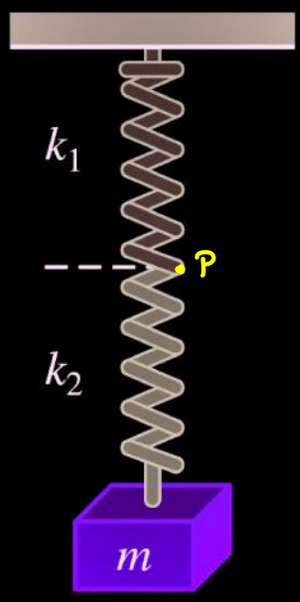
EFFECTIVE CONSTANT AND ELONGATION SHARE FOR SPRINGS IN SERIES:

Let x_1 and x_2 be the individual elongations in two springs connected in series and carrying a load F then considering FBD of junction (massless):

$$k_1 x_1 = k_2 x_2 = F \quad (1)$$

Also $x_1 + x_2 = x$ (2) where x is net elongation

From (1) & (2): $\frac{F}{k_1} + \frac{F}{k_2} = x = \frac{F}{k_{eff.}}$ (3)



(Where k_{eff} is the effective spring constant i.e constant of a spring which generates same F for same x ∴ $k_{eff}\, x = F$).

Thus:

$$\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} + \cdots$$

→ Individual elongations are inversely proportional to stiffness constants thus individual elongations can be found by dividing total

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elongation in inverse ratio of stiffness constants. Thus:

$$x_1 = \frac{1/k_1}{1/k_1 + 1/k_2}\, x = \frac{k_2}{k_1 + k_2}\, x \quad \text{or} \quad \boxed{x_1 = \frac{k_2}{k_1 + k_2}\, x}$$

Similarly $x_2 = \frac{k_1}{k_1 + k_2}\, x$.

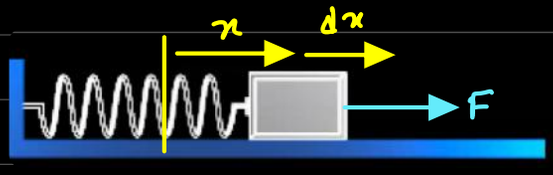
Stiffness Constant Of Portion Of A Spring:

Suppose a spring is divided in to n parts and we wish to find stiffness constant of each portion, then we can visualise the whole spring as series combination of the n partial springs. Now let k_n be the spring constant of the nth fraction of the whole spring, then by our series formula:

$$\frac{1}{k_n} + \frac{1}{k_n} + \dots \text{ } n \text{ terms} = \frac{1}{k} \Rightarrow \frac{n}{k_n} = \frac{1}{k}$$

$\Rightarrow \boxed{k_n = nk}$ thus if a spring of constant k is divided in to two parts, each parts has a constant $2k$. The same idea can also be extended to arbitrary fraction of the spring. For instance, spring constant of $2/3$ of a spring will be $3/2\,k$. (It can be thought as series combination of two springs of $1/3$ length).

Work Done In Stretching A Spring:



Suppose a spring is being gradually stretched. Let it be stretched through a distance x and let dx be the further stretch, then for gradual movement, $F = kx$ accordingly elemental work done in the process:

$dW = kx\,dx$, thus total work done:

$$W = \int_0^x kx\,dx = \frac{1}{2}kx^2$$ (Final elongation is represented by x here).

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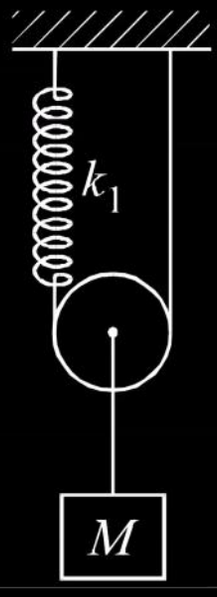
This work is stored as potential energy in the spring about which you will get more insights in the chapter on work power and energy. Thus for springs, stored potential Energy:

$$U = \frac{1}{2} k x^2$$

Where x is the deformation (either compression or elongation).

→ This energy can be retrieved back when spring returns to natural length.

Effective Spring Constant Of Spring Pulley Arrangements With Single Load:



When a gradually increasing load is applied to a system of spring pulley arrangement, To displace the system through a distance x, a gradually increasing force will be required whose final value is say F. Then the effective spring constant is simply defined as $k_{eff} = F/x$. The effective constant can thus be calculated using following ideas:

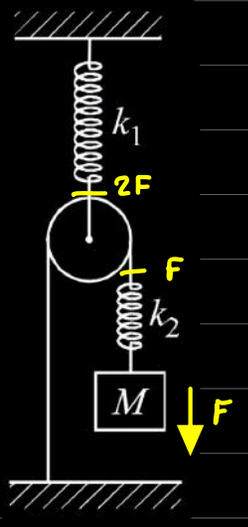
→ For massless pulleys, net force is zero, this idea can be used to find tensions in various branches containing springs.

→ Once the spring tensions are known, individual elongations in springs can be found by dividing Tensions by respective spring constants

→ Now total displacement of the load point can be found using visual inspection and then we can calculate $k_{eff} = F/k$.

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EXAMPLE: Find effective spring constant of the system shown in adjacent figure, using the method described in the previous section.



Solution: Instead of M imagine a gradually increasing force whose final value is F. Then tensions in the springs must be F and 2F as shown and accordingly their elongations must be $x_2 = F/k_2$ and $x_1 = 2F/k_1$ for the k_2 and k_1 springs respectively. Thus by visual inspection, load point must descend by an amount $x = 2x_1 + x_2 = \frac{4F}{k_1} + \frac{F}{k_2}$

Thus now $k_{eff} = \frac{F}{x} = \left[\frac{4}{k_1} + \frac{1}{k_2}\right]^{-1}$ Ans.

If we consider $f_1 = 2$ as force magnification factor for spring k_1 and $f_2 = 1$ for k_2 then, $k_{eff} = \left[\frac{f_1^2}{k_1} + \frac{f_2^2}{k_2}\right]^{-1}$.

This expression is generally true and can be understood in terms of work and energy principle.

FINDING EFFECTIVE SPRING CONSTANT USING WORK ENERGY PRINCIPLE:

Let a linearly increasing force cause a displacement of the loading point by a distance x. Then work done by this force:

$W = \frac{1}{2} Fx$ $\left(\because F = k_{eff} \times x,\ W = \frac{1}{2} k_{eff} x^2 = \frac{1}{2} Fx\right)$.

Now let the system have several springs to store the energy provided by this force, further let f_i be the magnification factor for i^{th} spring, then elongation in i^{th} spring, $x_i = \frac{f_i F}{k_i} \Rightarrow$ energy stored in i^{th} spring is $\frac{1}{2} k_i \left(\frac{f_i F}{k_i}\right)^2 = \frac{f_i^2 F^2}{2k_i}$, thus by work-energy principle,

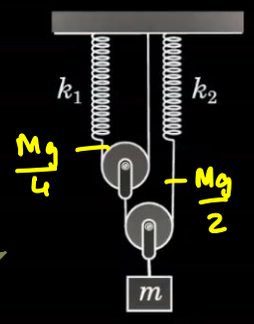
$$\frac{1}{2} Fx = \frac{F^2}{2} \sum \frac{f_i^2}{k_i} \Rightarrow k_{eff} = \frac{F}{x} = \left[\sum \frac{f_i^2}{k_i}\right]^{-1}$$

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Thus $\boxed{k_{eff} = \left[\sum \frac{f_i^2}{k_i}\right]^{-1}}$ Where f_i is force magnification factor i.e multiple of the load which is the tension in a given spring.

EXAMPLE: PATHFINDER: MCQ 34:

In the arrangement shown, the springs are light and have stiffness k_1 =100 N/m and k_2 = 200 N/m and the pulleys are ideal. An 8.0 kg block suspended from the lower pulley is initially held at rest maintaining the strings straight and springs relaxed. Now the force supporting the block is gradually reduced to zero. How far does the block descend during the process of reduction of the force? Acceleration due to gravity is 10 m/s².



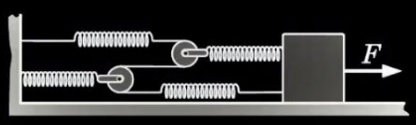
(a) 10 cm (b) 15 cm
(c) 1.6 m (d) None of these

Solution: The magnification factors f_1 and f_2 are 1/4 and 1/2 respectively for k_1 and k_2 Thus, $k_{eff} = \left[\frac{1}{16k_1} + \frac{1}{4k_2}\right]^{-1} = \left[\frac{1}{1600} + \frac{1}{800}\right]^{-1}$

$\Rightarrow k_{eff} = 1600/3$ N/m. Now $x = Mg/k_{eff} = 80/(1600/3) = 3/20$ m $= 15$ cm Ans.

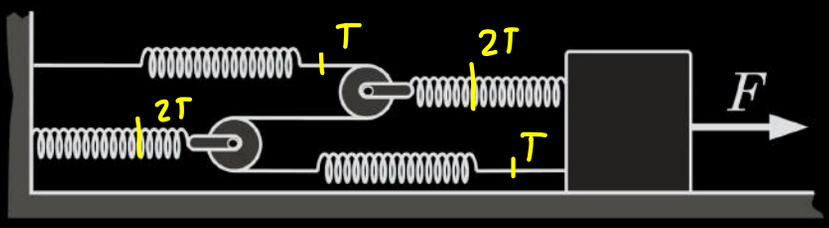
EXAMPLE: PATHFINDER: MCQ 35:

In the setup shown, a block is placed on a frictionless floor, the cords and pulleys are ideal and each spring has stiffness k. The block is pulled away from the wall. How far will the block shift, while the pulling force is increased gradually from zero to a value F?



(a) $\frac{2F}{5k}$ (b) $\frac{10F}{3k}$
(c) $\frac{8F}{9k}$ (d) $\frac{10F}{9k}$

Solution: Here we do not know tensions in the springs directly however we will soon find out. We assume a tension T in the lower spring connecting the block and other tensions are figured out in terms of T using the fact the pulleys (and springs) are massless. Now by equilibrium of block:



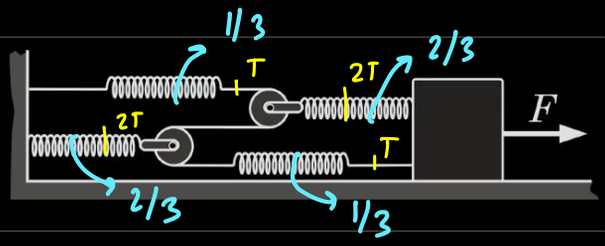
$F = 3T \Rightarrow T = F/3$. Thus now we have magnifying factors for each spring as shown in the figure below:

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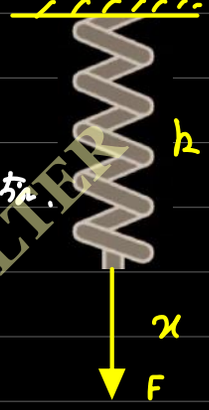
Now $k_{eff} = \left[\frac{1}{9k} + \frac{1}{9k} + \frac{4}{9k} + \frac{4}{9k}\right]^{-1}$

$\Rightarrow k_{eff} = 9k/10$. Thus required shift:

$x = F/k_{eff} = 10F/9k$ Ans. (d).



Minimum Constant Force To Pull A Relaxed Spring Through A Distance x:



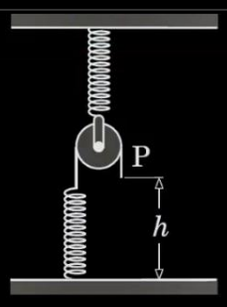
Instead of a gradually increasing force suppose we were to apply a constant force and we wish to know maximum elongation of a spring from such a configuration. Once again we can use work-energy principle here. Let max elongation be x then:

$$Fx = \frac{1}{2}kx^2 \Rightarrow \boxed{x = 2F/k}.$$

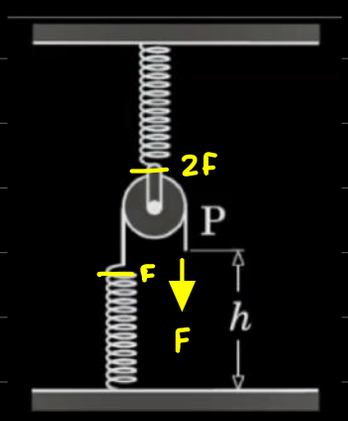
(Thus maximum elongation is double the equilibrium elongation, $\because$ at equilibrium, $F = kx$ or $x = F/k$]. Alternatively we can also infer that minimum constant force to elongate the spring by distance x is $kx/2$.

Example: Pathfinder: Build: 13:

A light frictionless pulley is suspended with the help of a light spring. One end P of a light inextensible thread that passes over the pulley is free and the other end is tied to another light spring that is affixed to the ground at its lower end as shown in the figure. Stiffness of both the springs is $k = 500$ N/m. The free end P is $h = 10.0$ cm above the ground. What minimum pull at the end P will bring it to the ground?



Solution: The magnifying factors are 2 and 1 as evident from the adjacent figure. Thus, $k_{eff} = \left(\frac{4}{k} + \frac{1}{k}\right)^{-1}$



i.e $k_{eff} = k/5$. Now by work energy principle,

$$\frac{1}{2}\left(\frac{k}{5}\right)h^2 = Fh \Rightarrow F = \frac{kh}{10} = \frac{500 \times 0.1}{10} = 5N$$

[Pathfinder Answer is 10N which is not correct. That would have been the case if a gradually increasing force were applied. Here the maximum extension position is not the equilibrium position. If we wanted equilibrium at ground, then our answer

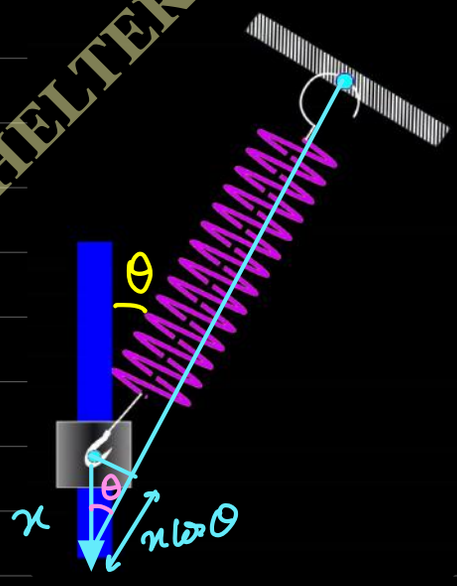
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could be calculated as $k_{eff} \times h = F = \frac{k}{5} h = 10N$, however that's not what the question says).

→ Caution: While using effective spring method, ends of the spring should move along the spring only, if they are moving at some angle then F does not vary linearly with x and in general we cannot find effective spring constant for such cases.

Effective Constant Of Spring Mass Collar System (Small Displacements):

Consider a spring mass collar arrangement in equilibrium as shown in the figure. Now if the mass is pulled down additionally by a small distance x, the spring further elongates by a distance $x\cos(\theta)$. Thus additional spring force is $F = kx\cos\theta$, however due to the collar's constraint it is free to move only in vertical direction, thus effective restoring force is $F_{eff} = kx\cos\theta(\cos\theta) = kx\cos^2\theta = k\cos^2\theta \times x$.

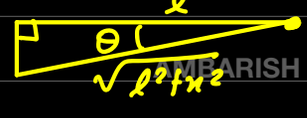


Thus the spring is behaving as if it's effective spring constant were $\boxed{k_e = k\cos^2\theta}$. This result becomes very useful in Simple Harmonic motion problem. For a vertically oscillating spring mass system, time period of oscillation is $T = 2\pi\sqrt{m/k}$. Thus for this system, time period can be readily calculated as $\boxed{T = 2\pi\sqrt{m/k_e}}$

→ This cannot be used if θ varies significantly. In that case the restoring force will not be proportional to x.

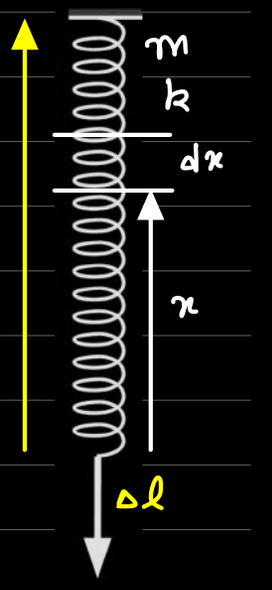
→ This cannot be used with $\theta = 90°$. In that case the restoring force can be obtained using pythagoras theorem and binomial approximation. The force in such a case is proportional to x^3 instead of x^1.

$$F = k[\sqrt{x^2+l^2} - l]\cdot\sin\theta \approx \frac{kx^2}{2l}\cdot\frac{x}{l} = \frac{kx^3}{2l^2}$$



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ELONGATION IN MASSIVE SPRING DUE TO ITS OWN WEIGHT:



Suppose we have a uniform spring of mass m and a natural length l (say when placed on smooth horizontal surface). We wish to know its elongation under its own weight. We will try to find the total elongation by integrating the elemental elongations. Consider an element of natural thickness dx at a natural coordinate x from the bottom. The tension in this element must be $T = \frac{x}{l} mg$ (∵ fractional weight supported by it) and its spring constant is $k' = k/(dx/l)$ (∵ fractional length is dx/l). Recall that spring constant of half spring was $2k$, thus smaller the portion, larger the spring constant. Now elongation in the element is simply $dy = T/k' = \frac{x\,mg}{l} \times \frac{dx}{lk} = \frac{mg}{kl^2}\,x\,dx$

Now to get total elongation, we need to add up all the elemental elongations. Thus:

$$\Delta l = \frac{mg}{kl^2}\int_0^l x\,dx = \frac{mg}{2k}.$$ Thus for purpose of calculating elongation due to its own mass, a mass $m/2$ can be assumed at the end of the spring.

→ If in addition to its own mass there is an extra load Mg at the end then an additional extension of Mg/k will also be present just as in case of massless springs).

→ If one end of spring is fixed and the other end pulled by a speed v then kinetic energy of spring can be shown to be $1/6\,mv^2$ by integration method (assuming linear variation of velocity). This is as if a load $m/3$ were attached at the end. (This is considered as effective mass in SHM problems).

→ In a massless uniform spring, elongation in any part is proportional to the length of that portion (Tension is same throughout).

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Example: Pathfinder: MCQ 12:

Length of a one-metre long uniform spring of mass 50.0 g increases by 2.00 cm due to its own weight, if it is suspended from a fixed support. How much load should be suspended from the lower end of this spring, so that total extension becomes 10.0 cm.

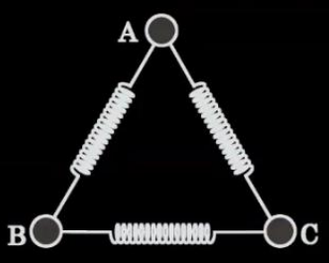
(a) 100 g
(b) 125 g
(c) 200 g
(d) Insufficient information

Solution: Let $m_1 = 50g$, $l_1 = 2\,cm$, $l_2 = 8\,cm$ thus according to our discussion from previous section,

$$k l_1 = \frac{m_1}{2} g \qquad k l_2 = m_2 g \quad \text{thus} \quad \frac{2m_2}{m_1} = \frac{l_2}{l_1}$$

$$\Rightarrow m_2 = \frac{1}{2} \times 50g \times \frac{8\,cm}{2\,cm} = 100\,g \quad \text{Ans (option a).}$$

Example: Pathfinder: MCQ 18:



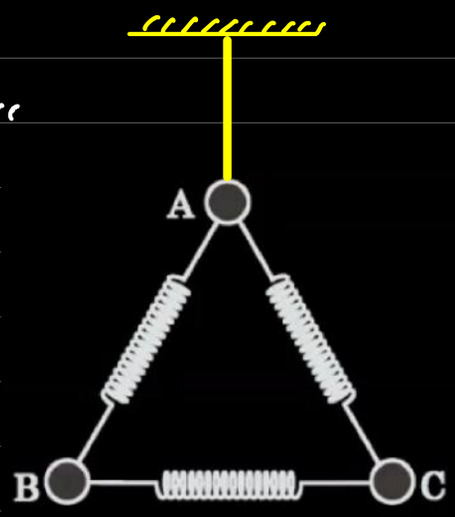
Three identical balls A, B and C each of mass m connected by three identical springs when placed on a frictionless horizontal floor occupy corners of an equilateral triangle as shown in the figure. This assembly is suspended by attaching the ball A to the ceiling with the help of a light thread. What can you predict regarding accelerations of the balls immediately after the thread is cut? Acceleration of free fall is g.

(a) All the balls will have downward acceleration g.
(b) All the balls have acceleration greater than g in different directions.
(c) Acceleration of the upper ball is $3g$ downwards and the accelerations of the lower balls are vanishingly small.
(d) Acceleration of the upper ball is $3g$ downwards and the lower balls move apart with finite accelerations.

Solution: **Concept:** Whenever a spring has massive objects / loads attached to its both ends, it cannot contract all of a sudden because that would require infinite acceleration of atleast one of the masses thus requiring infinite force. Thus if in any system involving cutting of strings / springs, the springs with masses on both ends have their tensions unchanged immediately after cutting of other springs / strings, while if due to cutting, masses / load at one of the ends of a spring is removed, it can contract all of a sudden (∵ a massless object can have infinite acceleration). Now let us

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apply these ideas to current problem:



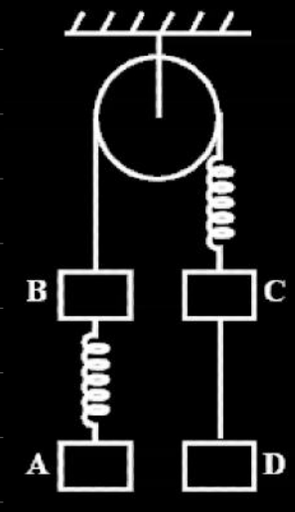
Prior to cutting of the string, resultant force on each ball is zero, and the tension in string is 3mg. Now when string is cut, the resultant spring force on each ball still remains the same, since none of the springs contracts suddenly. Thus balls B and C must have zero acceleration (No change in any force). Now for ball A, the net change in force is 3mg downwards which must also be the net force since the initial force was zero and other forces (spring force & gravity) are unaltered. Thus acc^n of A: $a_A = 3mg/m = 3g$ downwards. Thus correct option is C.

EXAMPLE: For the system shown in adjacent figure, all blocks are identical and pulley strings and springs are ideal. Find accelerations of all the blocks if:

(a) Spring between A and B is cut

(b) Spring above C is cut

(c) string between C and D is cut



Solution: (a) If the spring between A and B is cut, it contracts instantaneously while spring above C cannot contract suddenly due to loads at both ends. Thus forces on C and D remain unchanged and thus $a_C = a_D = 0$, $a_A = g$, Now force on B changes by mg upwards (∵ other forces are unchanged and downward spring tension of mg got removed). Thus $a_B = g$ upwards.

(b) When spring above C is cut, it contracts suddenly and thus C and D get in to free fall (consider combined FBD) thus $a_C = a_D = g \downarrow$ Now the force on A remains unchanged because the spring attached to it can't contract suddenly. Thus $a_A = 0$, while for B, the upward tension of 2mg has suddenly gone missing thus change in it's force is 2mg downwards, which must also be the net force on it.

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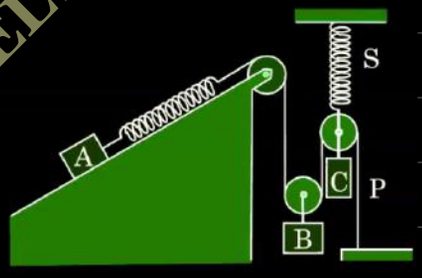
Thus, $a_B = 2mg/m \downarrow = 2g \downarrow$

(c) When string between C and D is cut, D gets in to free fall thus $a_D = g \downarrow$, Now none of the springs can contract suddenly, thus there is no change in force for A and B thus, $a_A = a_B = 0$ while for C, a downward tension of mg has gone missing thus change in force is mg upwards (other forces remain unchanged). Thus $a_c = mg/m = g \uparrow$.

EXAMPLE: PATHFINDER: BUILD:12:

The system shown in the figure is in equilibrium. The blocks A, B and C are of equal masses, the inclined face is frictionless, the pulleys are light and frictionless, the thread is light and inextensible, the springs are light and the blocks B and C are connected to the axles of the respective pulleys by rigid links. Find accelerations of the blocks A, B and C, immediately after

(a) the thread is cut at point P. (b) the spring S is cut.



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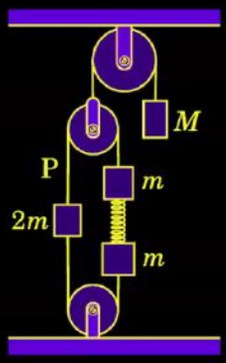
Solution: Initially tension in the spring on the incline is $mg/2$ (∵ Block B is in equilibrium) and that in S must be $2mg$ (∵ consider FBD of C and the pulley combined). Also angle of incline must be 30° ∵ $mg\sin\theta = mg/2$.

(a) If string is cut at point P, the spring on incline will contract suddenly, thus change in force on A is $mg/2$ down the incline, thus, $a_A = g/2 \swarrow$, B comes in free fall since the tension vanishes thus $a_B = g \downarrow$, now for C the change in force is mg upwards (string tensions vanish, other forces being the same) thus $a_c = mg/m$

$\Rightarrow a_c = g \uparrow$

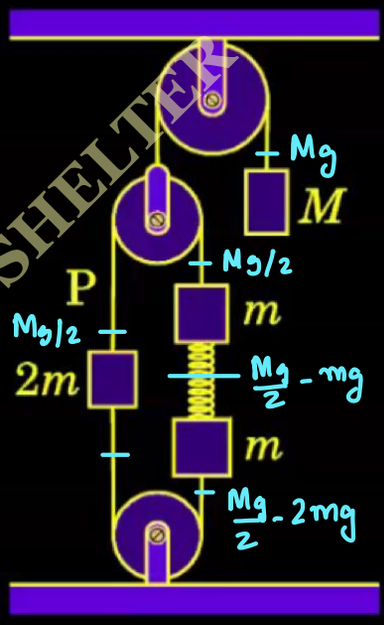
(b) When the spring S is cut, the spring on the incline cannot contract suddenly because that would require infinite acceleration of B on C, thus its tension (and therefore tension in string attached to it) remains unchanged. Thus for A and B there is no change in force ∴ $a_A = a_B = 0$, for C, the upward force of $2mg$ has vanished, so change in force is $2mg$ downwards (other forces are unchanged) thus $a_c = 2mg/m = 2g \downarrow$

Example: Pathfinder MCQ 19:



In the setup shown, the pulleys, the cords and the spring are ideal and masses of the loads are indicated in the figure. Initially the system is in equilibrium. What should be the range of mass M so that acceleration of the load of mass $2m$ becomes greater than acceleration of free fall immediately after the cord is cut at point P?

(a) $M > 4m$ (b) $M > 6m$
(c) $M > 8m$ (d) $M > 14m$



Solution: Using equilibrium considerations, the tensions in strings/springs are obtained as shown in adjacent figure. Upon cutting the cord at P, For 2m to fall with an accelenation greater than g, the string below it must have a non zero (positive) tension. For this, the lower block connected to the spring must have an upward accelenation greater than g, and equal to that of 2m. This means that if tension in the string below 2m were zero, the block below spring will have upward accelenation greater than g. Thus:

$$\left(\frac{Mg}{2} - mg - mg\right)/m > g \qquad (\because \text{FBD of block below spring assuming no } T).$$

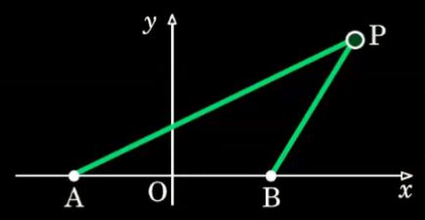
$$\Rightarrow \quad Mg > 6mg \Rightarrow \quad M > 6m$$

Point Spring: A spring with zero natural length is called a point spring. (Practically a spring with very small length in comparison to its elongation, may be treated as a point spring for some purposes. If the fixed end of such a spring is at $\vec{r}_0$ and the other end is pulled to $\vec{r}$ then the force vector offered by such a spring is $-k(\vec{r} - \vec{r}_0)$. This vector relation can be used for simplifying the problems involving point springs.

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Example: Pathfinder: Build 4:

Two identical elastic cords of negligible relaxed lengths are tied at one of their ends to fixed nails A and B that are equidistant from the origin O. The other ends of the strings are tied to a small ball. To hold the ball in equilibrium at a point P (4 m, 3 m), a force of magnitude $F = 1000$ N is required. Assuming free space conditions, find force constant of the cords?



Solution: Using theory discussed in previous section,

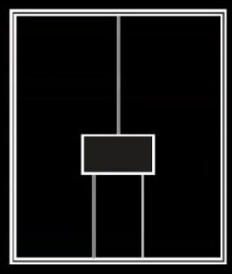
$$-k(\vec{r}_P - \vec{r}_A) - k(\vec{r}_P - \vec{r}_B) + \vec{F} = 0 \Rightarrow -k(2\vec{r}_P - (\vec{r}_A + \vec{r}_B)) = -\vec{F}$$

But $\vec{r}_A + \vec{r}_B = 0$ ($\because OA = OB$), thus $-2k\vec{r}_P = -\vec{F} \Rightarrow 2k r_P = F$

$\Rightarrow 2k \times 5 = 1000 \Rightarrow k = 100$ N/m Ans.

Example: Pathfinder: Check 7:

A block of mass $m = 15$ kg is suspended in an elevator with the help of three identical light elastic cords stretched vertically. One of them is tied to the ceiling and the other two are tied to the floor of the elevator as shown in the figure. When the elevator is stationary, the tensile force in each of the lower cords is $F = 7.5$ N. Acceleration due to gravity is $g = 10$ m/s^2.



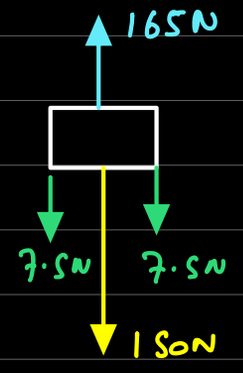
(a) For what acceleration of the lift, will the lower cords become relaxed?
(b) If the elevator moves with an upward acceleration $a = 1.0$ m/s^2, how much tensile force will develop in the upper cord?
(c) If the elevator moves with an upward acceleration $a = 2.0$ m/s^2, how much tensile force will develop in the upper cord?
(d) For what acceleration of the lift will the upper cord become relaxed?

Solution: Since the cords are identical, any displacement of the block will cause an equal change in tension in each cord so long as all of them are taut (i.e magnitude of change in force is less than 7.5 N).

(a) Now if the lower cords become relaxed (just relaxed), their tensions have decreased by 7.5 N each and thus tension in upper cords must have increased by 7.5 N, thus causing a net change of $\Delta F = 7.5\,N \times 3$ in the upward direction.
(Pull from the lower cords has reduced by 7.5 N each)

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Thus acceleration of block must change by DF/m upwards i.e $(7.5 \times 3/15) = 1.5\ ms^{-2}$ upwards also.



(b) Initially when the lift was stationary Now to cause an acceleration of $1\ ms^{-2}$ upwards, total resultant force must change by 15 N in upward direction thus force in each cord must change by 5N (reduction in lower cords and increase in upper cords). Thus tension in upper cord will become 165+5 i.e 170 N.

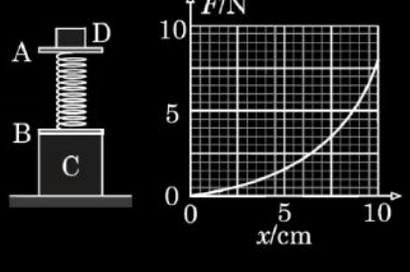
(c) If the elevator moves at $2\ ms^{-2}$ upwards lower cords must have already relaxed thus contributing a change of $7.5\ N \times 2$ i.e 15N in upward direction (recall that lower cords become relaxed at $1.5\ ms^{-2}$ itself). For an upward acceleration of $2\ ms^{-2}$, net change in upward force must be 15×2 i.e 30N out of which 15 N is contributed by lower cords so remaining 15 N must have been contributed by the upper cord thus tension in upper cord must be now 165+15 i.e 180 N also.

(d) For upper cord to just become relaxed, change in its tension must be 165 N and that must also be increase in tension of the lower cords thus net change in downward force must be 165×3 i.e 495 N. Thus acceleration will be 495/15 i.e $33\ ms^{-2}$ downwards also.

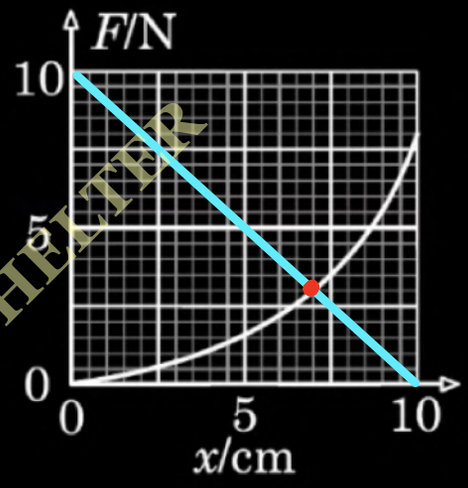
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EXAMPLE: PATHFINDER: BUILD 6:

Two light discs A and B are attached at the ends of a spring of force constant 100 N/m. The assembly is placed on a rubber pad C, which is placed on the floor. When a load D is placed on the disc A, the disc shifts downwards and stays there in equilibrium simultaneously the rubber pad is compressed and the disc B shifts downwards by a distance x. The restoring force F of the rubber pad varies with its compression x according to the given graph. Acceleration due to gravity is 10 m/s². Find mass of the load, so that when equilibrium is established, the disc A shifts 10 cm downwards.



Solution: If disc A shifts 10 cm downwards, l/s rubber pad be compressed a distance x cm thus compressive force in spring, F = 1N/cm x (10 - x) cm i.e (10 - x) N. This must also be the compressive force in the rubber pad because disc B is light. Thus to find exact compressive force, we need to locate a point on the curve where force satisfies F = 10 - x i.e intersection of line F = 10 - x with the given curve. Such a line and its intersection point are shown in the adjacent figure. The required F is thus 3N, thus mass of the load must by 300 gm Ans.
(Corresponding compression in rubber pad is 7cm).



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